

Spectral Theory of Graphs

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Primary References

1. *Spectral and Algebraic Graph Theory* by Daniel Spielman
2. *Spectral Graph Theory and Related Topics Lectures* by Matthew Hirn

Basic Definitions

Definition 1. A *Graph* is a pair (E, V) such that V is a set of vertices and $E \subseteq \binom{V}{2}$. Note that this definition disallows loops and multiple edges. A *Weighted Graph* is a triple (E, V, f) where (E, V) is a graph and $f : E \rightarrow \mathbb{R}$. We will primarily use weights in $\mathbb{R}$.

Definition 2. The adjacency matrix of a finite graph is

$$\mathbf{M}_G(a, b) = \begin{cases} 1, & (a, b) \in E \\ 0, & (a, b) \notin E \end{cases}$$

We can also modify this by changing $1 \rightarrow w(a, b)$ for a weighted graph.

Definition 3. For a vertex a in a graph (E, V) , the neighborhood of a is the set of all vertexes connected to a by an edge.

$$N_G(a) = \{b \in V : (a, b) \in E\}.$$

The degree of a vertex a is

$$\deg(a) = |N_G(a)|.$$

If it is a weighted graph, we add up all the weights of the incident edges.

Definition 4. The degree vector $\mathbf{d}$ is given by

$$\mathbf{M1}.$$

Definition 5. The degree matrix of a graph is obtained by placing the degrees in a diagonal matrix.

$$\mathbf{D}_G(a, a) = \mathbf{d}(a)$$

and 0 on all off diagonals.

Definition 6. The random walk matrix of a graph is

$$\mathbf{W}_G = \mathbf{M}_G \mathbf{D}_G^{-1}$$

The Graph Laplacian

The normal laplacian $\Delta f = \sum_{k=1}^n \frac{\partial^2 f}{\partial x_k^2}$. In one dimension, this is simply $\Delta f = f''$.

Lemma 7. Note that

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c-h)}{2h}.$$

Proof Sketch: Add a middle term $f(c) - f(c)$ and obtain two copies of the definition of derivative (one with h replaced with $-h$).

Lemma 8. Observe that

$$f''(c) = \Delta f(c) = \lim_{h \rightarrow 0} \frac{f(c+h) + f(c-h) - 2f(c)}{h^2}$$

Proof Sketch: Use L'Hospital's rule to turn it into previous lemma.

Definition 9. The graph Laplacian of a graph G is given by

$$L_G = D_G - M_g$$

Note that for a simple graph, this matrix has the degrees on the diagonal and a -1 whenever two vertices are connected.

Lemma 10. For any vector $x \in \mathbb{R}^{|V|}$, we have

$$x^T L x = \sum_{(a,b) \in E} (x(a) - x(b))^2$$

Proof:

$$(Lx)_a = \sum_{b \in V} L_{ab} x(b) = d(a)x(a) + \sum_{b \neq a} -\delta_{ab} x(b)$$